London, July 10, 2026

Re: RL Fellows Program

To whom it may concern at *Anthropic*,

I am writing to apply for the **RL Fellows Program** at **Anthropic**. **My research** sits at the intersection of deep RL and large language models, with a focus on **credit assignment** and building **scalable RL environments**. I combine an academic track record in RL/LLMs with five years of industry experience shipping ML systems, and I am motivated by the challenge of **scaling RL** to produce safe, capable AI — which is precisely what this Fellowship is designed to accelerate.

Academic and research experience. Currently, I am an **Assistant Professor in Machine Learning at UCL** and **Associate Faculty** at the *BOLD Lab* — a £60M UKRI-funded national AI research lab unifying Oxford, UCL, and Imperial around open-ended learning, embodied AI, and new learning paradigms beyond backpropagation, led by Jakob Foerster. At UCL, I lead three modules: *ELEC0136: Machine Learning System Design*, *ELEC0021: Programming for Machine Learning*, and *ELEC0054: Research Projects*, where I coordinate UCL's thesis-project portfolio. I also teach in *ELEC0130: Internet of Things*, *ELEC0149: Introduction to Machine Learning*, and *ELEC0135: Deep Learning*, and contribute to syllabus design for Deep Learning and NLP. I was nominated for the **UCL Inspiring Teaching Delivery award** two years in a row (2025 and 2026).

Before this, I completed my PhD at **UCL** under the supervision of Laura Toni and Tim Rocktäschel at the intersection of **Deep Reinforcement Learning** (RL) and **Large Language Models** (LLMs) with a focus on the Credit Assignment Problem. I reviewed the field with *A Survey on the Temporal Credit Assignment Problem*, which received the TMLR "Survey Certification" **award** for *exceptional thoroughness*. I showed that **Large Language Models** align with human evaluations of action relevance across navigation tasks, and I developed **NAVIX**, a **widely adopted** JAX reimplementation of MiniGrid that enables faster iteration and cleaner evaluations. Most recently, I co-authored **BALROG** (ICLR 2025), a benchmark for evaluating LLM and VLM agents on long-horizon interactive tasks.

Research agenda. I care about **scaling RL** to real-world problems through principled environment design and credit assignment. A key conclusion from my doctoral studies is that effective RL training demands two things: **environments that expose long-horizon structure**, and **training schemes that propagate credit across it**. My current work targets this directly: I am developing **latent option models** that scale **options** — temporally extended actions — to large, complex environments. By learning a reusable vocabulary of latent actions from large-scale, **action-free** demonstrations, agents acquire a discrete action space that can then be fine-tuned with RLHF/RLAIF. This connects directly to the Fellowship's workstreams: multi-turn reasoning and long-horizon tool use in Claude require RL signals that propagate meaningfully across hundreds of steps — the harder the credit assignment problem, the more principled the environment and training scheme need to be.

Industry experience. Before academia, I spent **five** years translating early machine learning research into commercial products. At Burohappold (2017 to 2020, London, UK), as **Machine Learning Lead**, I delivered deep-learning tools now used by the **Premier League** and internal teams, creating new revenue streams and increasing operational efficiency. At Gridshell.it (2015 to 2017, Naples, Italy) I developed open-source multi-objective **genetic algorithms** for structural and acoustic engineering in resource-constrained settings; this work was recognised with the **Peter Lord Award**.

These roles developed my ability to align scientists, engineers, and stakeholders around ambitious, complex deliverables, and strengthened my **systems** mindset—building reproducible codebases and infrastructure in **JAX**/**PyTorch** for long-running experiments.

Why me? I have a rare combination of **RL research depth**, hands-on **JAX environment-building** experience, and a proven ability to **ship production ML systems**. My work on NAVIX shows I can build robust, scalable RL infrastructure: thousands of parallel environments on a single GPU, adopted across multiple research groups. My credit assignment research shows I understand the hard theoretical problems that sit beneath effective RL training. **My agenda**—principled environment design and temporally extended action learning to scale RL—**aligns** closely with the Fellowship's goal of improving Claude's capabilities through RL.

Five years of **industry experience** gave me a product-oriented, demand-driven mindset and the organisation and management skills to **plan effectively** and **execute efficiently**. My **working ethos** is grounded in teamwork, collaboration, and mutual understanding.

The **transition from architecture** to AI was a **proactive** leap that shows **flexibility**, rapid skill acquisition beyond my comfort zone, and sustained **determination** and **dedication**. In short, this multidisciplinary journey equipped me to manage and execute **complex**, **ambitious** projects that span multiple fields.

Why the RL Fellows Program at Anthropic? Anthropic is doing the most **ambitious RL-for-alignment** work in the field, at a scale and pace that no academic group can match. The Fellowship offers direct mentorship from researchers defining the frontier of post-training, access to compute that would compress years of progress into four months, and a **clear path** — 25–50% of Fellows receive full-time offers — to contributing to that mission long-term. I thrive where the challenge is not between a bad and a good idea, but between a **good** and a **great** one — settings that keep my motivation high and maximise the impact I can deliver.

Sincerely,
Eduardo Pignatelli

